Problem Formulation: Self-Driving Lab Simulator

Project: Autonomous Synthesis Optimization using Reinforcement Learning
Algorithm: REINFORCE (Monte Carlo Policy Gradient)

1. Environment Overview

The environment is a simulated multi-step chemical reactor formulated as a Continuous Markov Decision Process (MDP). The objective of the autonomous agent is to discover the optimal reaction parameters that maximize the synthesis yield of a target chemical product. To mimic the constraints of a physical laboratory, each experimental campaign (episode) is restricted to a finite budget of 25 sequential steps. The underlying chemical response surface is non-linear, and sensor readings are perturbed by Gaussian noise to simulate realistic measurement uncertainty.

2. State Space (S)

The state space is continuous, defined as a 3D vector $\mathbf{s_t} \in \mathbb{R}^3$ that tracks the current physical parameters of the reactor alongside the remaining experimental budget. The state vector at step *t* is defined as:

- **Normalized Temperature (T_t):** The current reactor temperature, physically bounded between 20°C and 150°C, starting at an ambient condition of 25°C. It is normalized by a factor of 150.0 for neural network stability.
- **Normalized Concentration (C_t):** The current reagent concentration, physically bounded between 0.1 M and 2.0 M, starting at a dilute condition of 0.5 M. It is normalized by a factor of 2.0.
- **Remaining Budget Fraction:** A scalar representing the ratio of remaining steps to the total budget (25 steps), descending from 1.0 to 0.0. This makes the MDP time-aware.

3. Action Space (A)

The action space is continuous, allowing for fine-grained adjustments to the synthesis conditions. The agent outputs a 2D continuous action vector $\mathbf{a_t} = [\Delta T, \Delta C] \in [-1, 1]^2$. These normalized actions are scaled into physical adjustments:

- **Temperature Adjustment (ΔT):** Scaled to a maximum adjustment of ±20.0°C per step.
- **Concentration Adjustment (ΔC):** Scaled to a maximum adjustment of ±0.5 M per step.

Post-adjustment, the new state variables T_{t+1} and C_{t+1} are strictly clamped to their respective physical bounds to prevent the simulation of impossible chemical states.

4. Reward Function (R)

The reward function is formulated to promote high absolute yield while discouraging wasteful exploration. The reward at step *t* is defined as $\mathbf{R_t = Y_{obs} + Bonus_{discovery} - Penalty_{reagent}}$, where:

- **Observed Yield (Y_{obs}):** A bivariate Gaussian response surface peaking at 1.0 (100% yield) when T = 85.0°C and C = 1.20 M. Real-world sensor noise is modeled by adding $\varepsilon \sim N(0, 0.02^2)$ to the true yield, clamped between 0 and 1.
- **Discovery Bonus:** To incentivize breakthrough discoveries, a sparse bonus of $2.0 \times (Y_{obs} - Y_{best})$ is awarded only when the current yield exceeds the maximum yield previously observed in the episode.
- **Reagent Penalty:** A linear cost penalty proportional to the current concentration ($0.1 \times C_t$), representing the financial and material cost of consuming excess chemical reagents.